- Acceleration of RSA, DH, and ECC over GF(p) operations, based on the Montgomery method for fast modular multiplications. More specifically:
  - RSA modular exponentiation, RSA Chinese remainder theorem (CRT) exponentiation
  - ECC scalar multiplication, point on curve check, complete addition, double base ladder, projective to affine
  - ECDSA signature generation and verification
- Capability to handle operands up to 4160 bits for RSA/DH and 640 bits for ECC
- When manipulating secrets: protection against side-channel attacks (SCA), including differential power analysis (DPA), certified SESIP, and PSA security assurance level 3
  - Applicable to modular exponentiation, ECC scalar multiplication, and ECDSA signature generation
- Arithmetic and modular operations such as addition, subtraction, multiplication, modular reduction, modular inversion, comparison, and Montgomery multiplication
- Built-in Montgomery domain inward and outward transformations
- AMBA AHB slave peripheral, accessible through 32-bit word single accesses only (otherwise an AHB bus error is generated, and write accesses are ignored)
- Support for CCB chaining operations required to protect the private key used in PKA protected operations
- Hardware protections to monitor the usage of private keys during protected operation initialization

3.26 Timers and watchdogs

The devices include two advanced control timers, up to eleven general-purpose timers, two basic timers, two low-power timers, two watchdog timers, and one SysTick timer.

The table below compares the features of the advanced control, general-purpose and basic timers.

Table 7. Timer feature comparison

<table>
<tr><th>Timer type</th><th>Timer</th><th>Counter resolution</th><th>Counter type</th><th>Prescaler factor</th><th>DMA request generation</th><th>Capture / compare channels</th><th>Complementary outputs</th></tr>
<tr><td>Advanced control</td><td>TIM1, TIM8</td><td>16 bits</td><td>Up, down, Up/ down</td><td>Any integer between 1 and 65536</td><td>Yes</td><td>4</td><td>4</td></tr>
<tr><td rowspan="4">General-purpose</td><td>TIM2, TIM3, TIM4, TIM5</td><td>32 bits</td><td>Up, down, Up/ down</td><td rowspan="4">Any integer between 1 and 65536</td><td>Yes</td><td>4</td><td>No</td></tr>
<tr><td>TIM12</td><td>16 bits</td><td>Up</td><td>No</td><td>2</td><td>No</td></tr>
<tr><td>TIM15</td><td>16 bits</td><td>Up, down, Up/ down</td><td rowspan="2">Yes</td><td>2</td><td>1</td></tr>
<tr><td>TIM16, TIM17</td><td>16 bits</td><td>Up, down, Up/ down</td><td>1</td><td>No</td></tr>
<tr><td>Basic</td><td>TIM6, TIM7</td><td>16 bits</td><td>Up</td><td>Any integer between 1 and 65536</td><td>Yes</td><td>0</td><td>No</td></tr>
</table>

3.26.1 Advanced-control timers (TIM1/TIM8)

The advanced-control timers (TIM1/TIM8) consist of a 16-bit autoreload counter driven by a programmable prescaler.

They may be used for various purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

TIM1/TIM8 timer features include: